Import libraries

In [67]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, accuracy_score, confusion_matrix, classification_report
from sklearn.decomposition import PCA
from xgboost import XGBClassifier
import warnings
warnings.filterwarnings("ignore")
```

Load dataset

In [68]:
```
df = pd.read_csv(r"C:\Users\anupa\Desktop\Python Learning\Cluster Analysis\Wholesale customers data.csv")

print(df.shape)
df.head()
```

```
(440, 8)
```

Out[68]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	2	3	12669	9656	7561	214	2674	1338
1	2	3	7057	9810	9568	1762	3293	1776
2	2	3	6353	8808	7684	2405	3516	7844
3	1	3	13265	1196	4221	6404	507	1788
4	2	3	22615	5410	7198	3915	1777	5185

Missing value treatment

In [69]:
```
#Check missing values first

df.isnull().sum()
```

Out[69]:
```
Channel             0
Region              0
Fresh               0
Milk                0
Grocery             0
Frozen              0
Detergents_Paper    0
Delicassen          0
dtype: int64
```

In [70]:
```
#If missing values exist:

df = df.fillna(df.median(numeric_only=True))
```

Exploratory Data Analysis (EDA)

Summary statistics

In [71]:
```
df.describe()
```

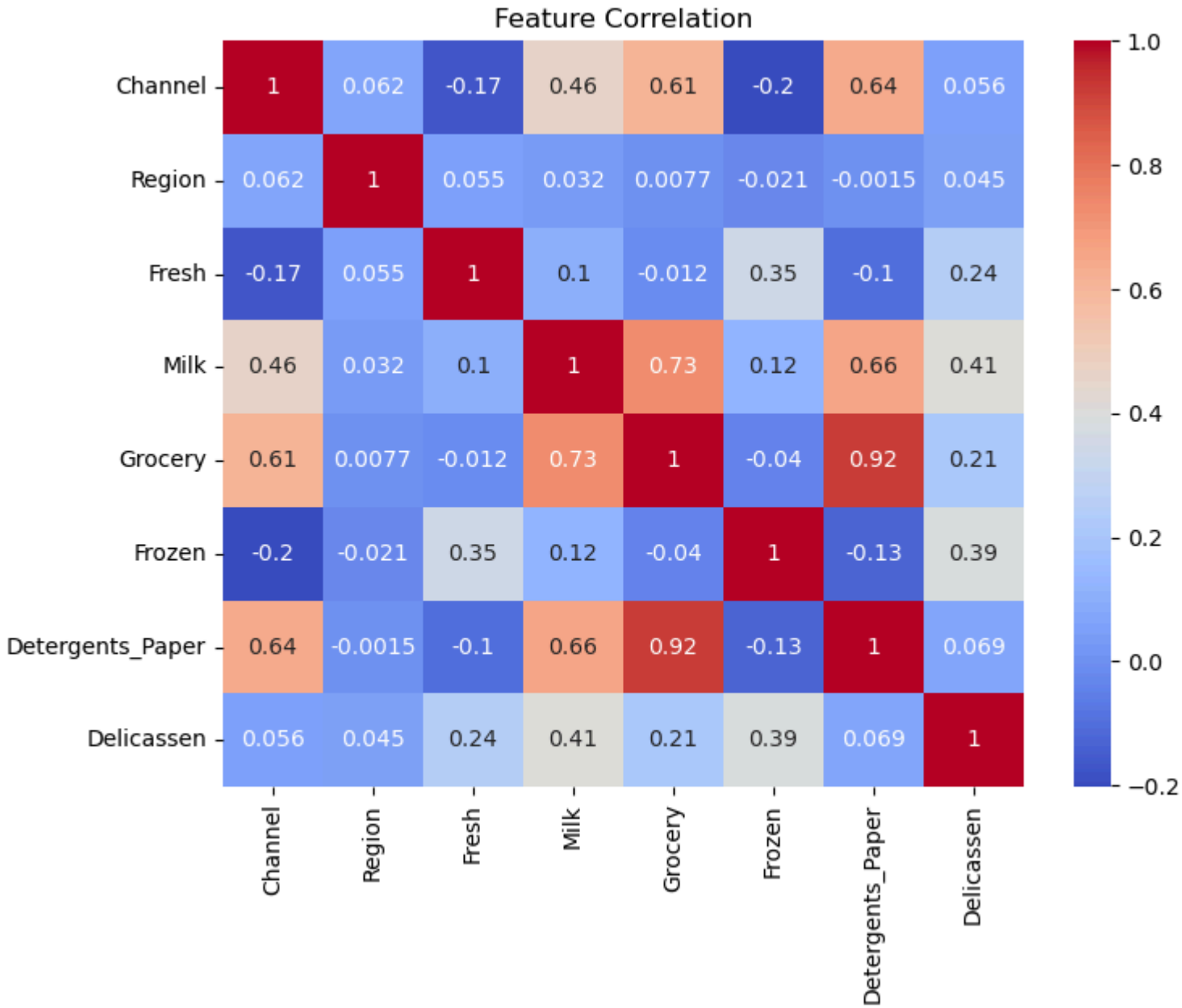
Out[71]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
count	440.000000	440.000000	440.000000	440.000000	440.000000	440.000000	440.000000	440.000000
mean	1.322727	2.543182	12000.297727	5796.265909	7951.277273	3071.931818	2881.493182	1524.870455
std	0.468052	0.774272	12647.328865	7380.377175	9503.162829	4854.673333	4767.854448	2820.105937
min	1.000000	1.000000	3.000000	55.000000	3.000000	25.000000	3.000000	3.000000
25%	1.000000	2.000000	3127.750000	1533.000000	2153.000000	742.250000	256.750000	408.250000
50%	1.000000	3.000000	8504.000000	3627.000000	4755.500000	1526.000000	816.500000	965.500000
75%	2.000000	3.000000	16933.750000	7190.250000	10655.750000	3554.250000	3922.000000	1820.250000
max	2.000000	3.000000	112151.000000	73498.000000	92780.000000	60869.000000	40827.000000	47943.000000

Correlation heatmap

In [72]:

```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Feature Correlation")
plt.show()
```



Remove categorical columns

In [73]:
```
X = df.drop(['Channel','Region'], axis=1)
```

Outlier detection (IQR method)

In [74]:
```
Q1 = X.quantile(0.25)
Q3 = X.quantile(0.75)

IQR = Q3 - Q1

X = X[~((X < (Q1 - 1.5 * IQR)) | (X > (Q3 + 1.5 * IQR))).any(axis=1)]
```

In [75]:
```
X.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 332 entries, 0 to 439
Data columns (total 6 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Fresh             332 non-null    int64
 1   Milk              332 non-null    int64
 2   Grocery           332 non-null    int64
 3   Frozen            332 non-null    int64
 4   Detergents_Paper  332 non-null    int64
 5   Delicassen        332 non-null    int64
dtypes: int64(6)
memory usage: 18.2 KB
```

Train–test split

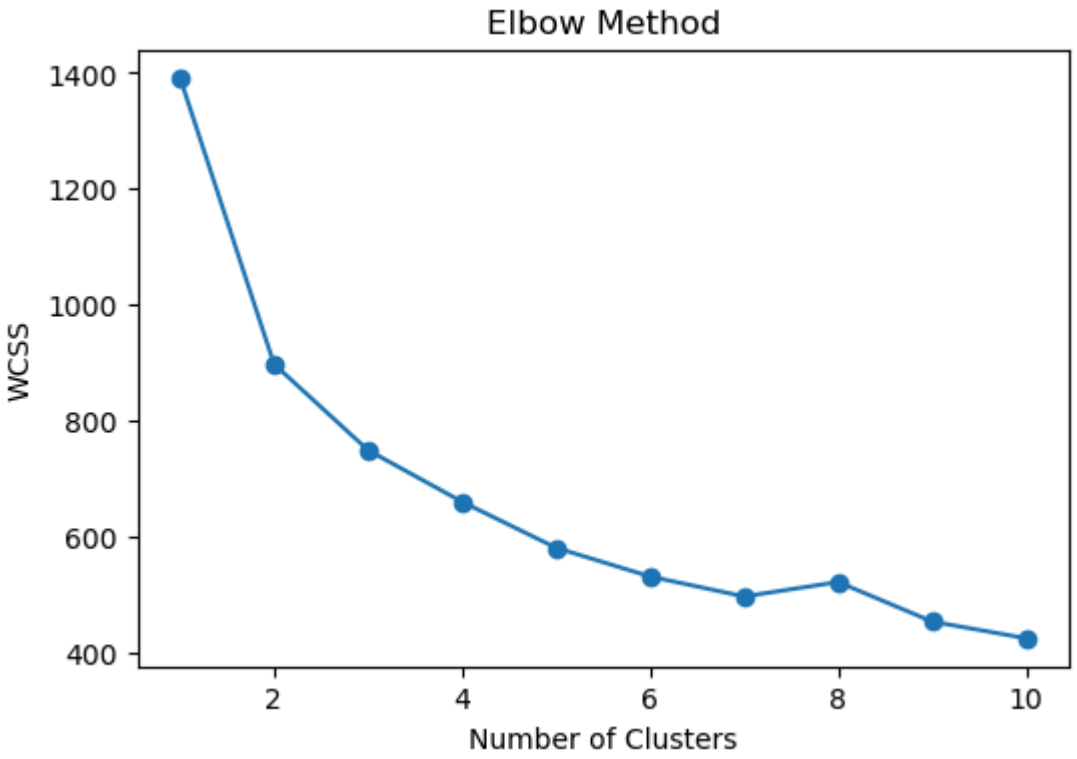
In [76]:
```
X_train, X_test = train_test_split(
    X,
    test_size=0.3,
    random_state=42
)
```

Feature scaling

In [77]:
```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Elbow Method (optimal clusters)

In [78]:
```
wcss = []

K_range = range(1,11)

for k in K_range:

    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_train_scaled)

    wcss.append(kmeans.inertia_)

plt.figure(figsize=(6,4))
plt.plot(K_range, wcss, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```



Train clustering model

In [79]:
```
kmeans = KMeans(n_clusters=3, random_state=42)

train_clusters = kmeans.fit_predict(X_train_scaled)
```

Cluster evaluation (Silhouette score)

In [80]:
```
sil_score = silhouette_score(X_train_scaled, train_clusters)

print("Silhouette Score:", sil_score)
```

```
Silhouette Score: 0.26452635168485633
```

Assign clusters to test data

In [81]:
```
test_clusters = kmeans.predict(X_test_scaled)
```

Train XGBoost classifier

In [82]:
```
model = XGBClassifier(
    n_estimators=200,
    max_depth=4,
    learning_rate=0.05,
    random_state=42
)

model.fit(X_train_scaled, train_clusters)
```

Out[82]:
```
▾ XGBClassifier
▶ Parameters
```

Predict clusters on test data

In [83]:
```
pred_clusters = model.predict(X_test_scaled)
```

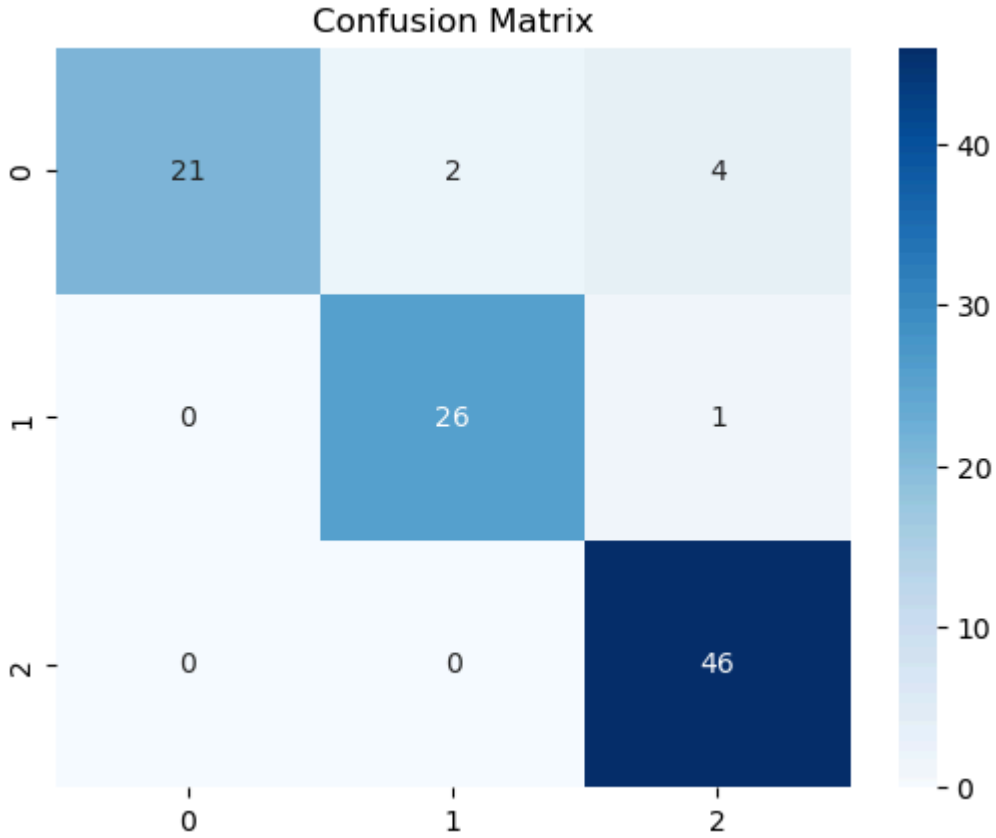
Model evaluation

In [84]:
```
accuracy = accuracy_score(test_clusters, pred_clusters)

print("Accuracy:", accuracy)
```

```
Accuracy: 0.93
```

Confusion matrix

In [85]:
```
cm = confusion_matrix(test_clusters, pred_clusters)

sns.heatmap(cm, annot=True, cmap='Blues')
plt.title("Confusion Matrix")
plt.show()
```



Classification report

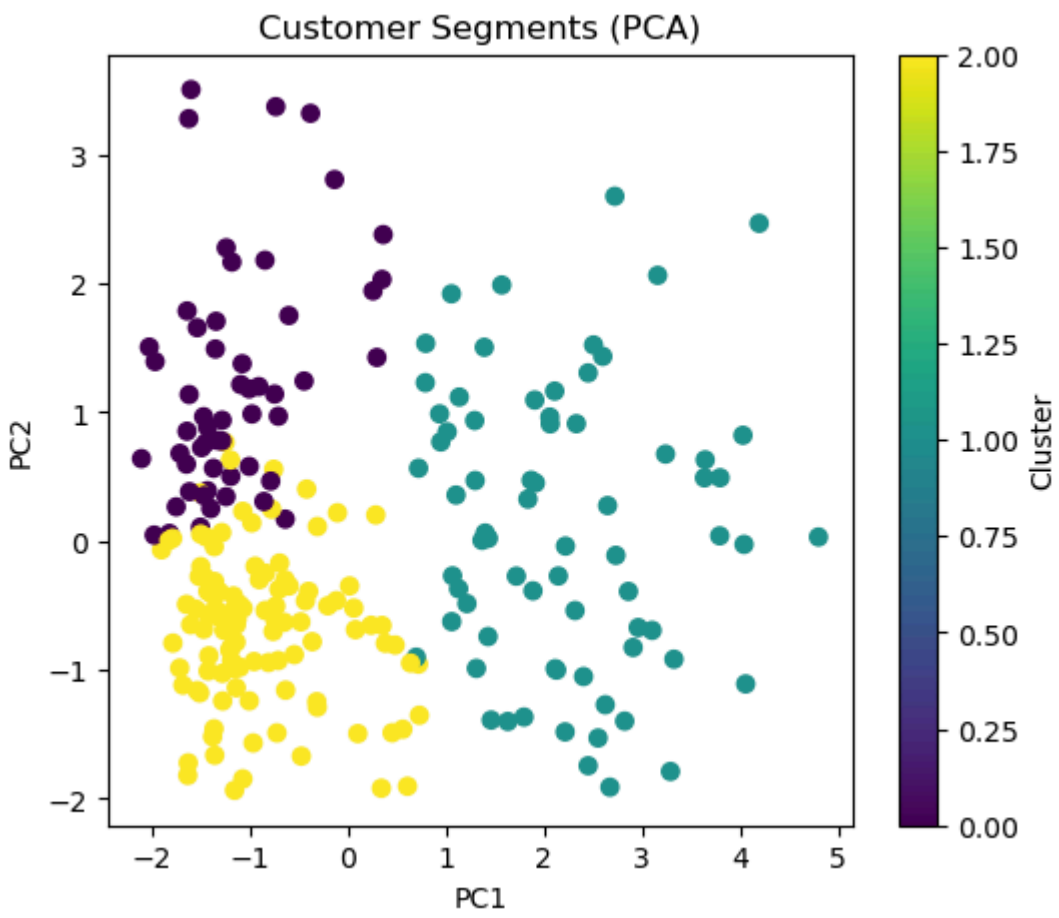
In [86]:
```
print(classification_report(test_clusters, pred_clusters))
```

```
              precision    recall  f1-score   support

           0       1.00      0.78      0.88        27
           1       0.93      0.96      0.95        27
           2       0.90      1.00      0.95        46

    accuracy                           0.93       100
   macro avg       0.94      0.91      0.92       100
weighted avg       0.94      0.93      0.93       100
```

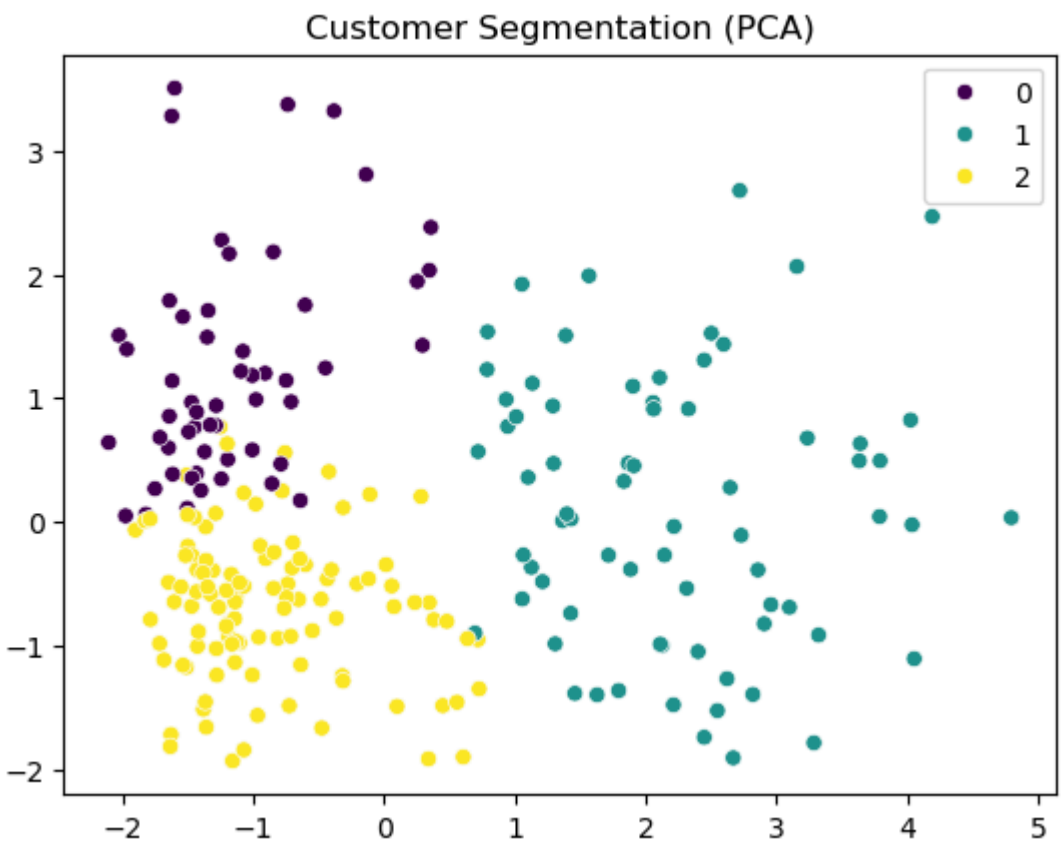
PCA cluster visualization

In [87]:

```
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

pca = PCA(n_components=2)

X_train_pca = pca.fit_transform(X_train_scaled)

plt.figure(figsize=(6,5))

plt.scatter(
    X_train_pca[:,0],
    X_train_pca[:,1],
    c=train_clusters,
    cmap='viridis'
)

plt.title("Customer Segments (PCA)")
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.colorbar(label='Cluster')
plt.show()
```



In [88]:
```
import seaborn as sns

sns.scatterplot(
    x=X_train_pca[:,0],
    y=X_train_pca[:,1],
    hue=train_clusters,
    palette='viridis'
)

plt.title("Customer Segmentation (PCA)")
plt.show()
```



Cluster interpretation

Attach clusters back to original data.

In [89]:
```
X_train_clustered = X_train.copy()

X_train_clustered['Cluster'] = train_clusters

cluster_summary = X_train_clustered.groupby('Cluster').mean()

print(cluster_summary)
```

```
                Fresh         Milk       Grocery       Frozen  \
Cluster
0        15570.518519  2309.240741   3270.462963  3925.092593
1         7710.746479  8107.929577  11834.408451  1439.239437
2         7656.542056  2518.607477   2996.242991  1234.897196

         Detergents_Paper   Delicassen
Cluster
0              477.888889  1209.685185
1             4844.267606  1427.422535
2              720.345794   616.504673
```

Business interpretation.

Example: Cluster 0 → Fresh product buyers Cluster 1 → Grocery retailers Cluster 2 → Balanced buyers